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Lightweight self-hosted AI chat web app. An Open WebUI alternative with **under 100 MB RAM**. No Docker. No build step. One Node.js process + one SQLite file.

Stack	Node.js + Express 5 + better-sqlite3 + vanilla JS
AI Backend	9Router (<code>http://localhost:20128/v1</code> , OpenAI-compatible)
Providers	AG, Groq, BAI, OpenRouter, Nvidia, Cline — 497+ models
Frontend	<code>index.html</code> + <code>style.css</code> + <code>app.js</code> , zero framework, dark theme
Language	English (default) / Indonesian — runtime toggle
Version	v1.0-beta
License	Private / personal use

Table of Contents

- 1. [Features](#)
- 2. [System Architecture](#)
- 3. [Request Flow Diagrams](#)
- 4. [Requirements](#)
- 5. [Installation](#)
- 6. [Management Script](#)
- 7. [Configuration](#)
- 8. [API Reference](#)
- 9. [Database Schema](#)
- 10. [Access from Other Devices](#)
- 11. [Performance](#)
- 12. [Security Notes](#)
- 13. [Troubleshooting](#)
- 14. [Changelog](#)

Features

Chat

Feature	Description
Real-time streaming	SSE relay, word-by-word from 9Router

Feature	Description
Multi-session history	Per-user isolation in SQLite
Chat memory	Token-budget context window (default 1600 tokens $\approx$ 15–30 msgs, oldest auto-compressed) + rolling background summary of older turns
Cross-chat memory	Auto-extracted personal facts per user (max 20), injected into every chat; editable in Settings → Memory
Token saver	<code>max_tokens</code> reply cap (default 1024), history budget, per-chat Lean ⚡ toggle (answers without history)
Per-reply usage badge	prompt / completion / total tokens shown under each AI reply
Rename / Delete chats	Via sidebar hover buttons; rename uses custom in-app dialog (bilingual, no native prompt)
Deep-link URLs	SPA router (History API): <code>/c/:id</code> lands in the address bar automatically when a reply finishes (ChatGPT/Gemini style; clicking a chat title also reveals it) · <code>/account</code> settings · <code>/admin</code> dashboard — refresh, bookmark & browser back all work (server catch-all → app; unknown <code>/api/*</code> stays JSON 404)
Share link (public)	Chat "... " → Share: read-only page at <code>/s/<token></code> (<code>noindex</code> , XSS-safe snapshot); revoke via warn modal; delete chat = revoke; empty chats blocked (<code>ERR_EMPTY_SHARE</code>)
Pin / Archive / Tag / Fork	Chat "... " menu: pin to top, archive to a collapsible "Archived (n)" section, inline #tag, fork = branch copy of the conversation (<code>fork</code>) suffix)
Chat search	Instant client-side filter by title (sidebar)
Markdown + code blocks	With streaming auto-close fence fix
Copy & Retry	Per-message actions
Smart chat title	After the first reply a short 3–6 word title is generated in the background (never overwrites a manual rename)
Edit & regenerate	Pencil on any user message → inline edit → save truncates everything after it; ↻ button regenerates from the last user message
Code highlight + copy	fenced blocks get language colouring (bundled hljs, no CDN) + per-block copy button
Stop generation	Abort mid-stream
Provider timeout	Admin-configurable 30–600 s (default 120 s), applies to all upstream calls
Typing indicator	Animated dots while waiting
Temporary Chat	Zero history saved (endpoint: <code>/api/temp-chat</code>); honours Lean mode (skip history + memory = zero context)
Realtime quota UI	Sidebar quota + admin dashboard refresh automatically after every reply (dashboard auto-polls every 5 s while open)

Users & Access

Feature	Description
Login	script per-user salt (<code>script1:salt:hash</code>), timing-safe compare; legacy static-salt hashes auto-upgrade on login
Session purge	Expired sessions deleted on every login
Roles	<code>admin</code> (full control) and <code>user</code> (chat only)
Daily quota	Per-user limit (default 50/day), admin unlimited; quota bar shown in sidebar with model chip
User management	Admin: add / delete / reset password / activate / deactivate (self-deletion & self-deactivation blocked server-side)
Instant session revoke	Deactivate or password reset kills sessions immediately
Quota edit & reset	Admin edits any user's daily quota (1–999999) and resets one or all users' usage
Reset password (admin)	Custom in-app modal with validation (min 4 chars, clear on short) — no native <code>prompt()</code> anywhere
Deactivate/Activate/Delete guard	Open WebUI-style warn modals with user name + consequence text (red danger for deactivate/delete, blue/info for activate); actions only run after confirm

Guest Mode (no login)

Feature	Description
Try as guest	Button under Login on the login screen (only shown when guest mode is enabled + gateway configured); clicks straight into the chat dashboard
Configurable limits	Admin sets max messages per guest (1–200, default 10) and session time limit (1–60 min, default 5)
Live guest chip	Sidebar shows <code>Guest — n/max messages · mm:ss left</code> (turns red under 60 s)
Login popup on exhaustion	When chats or time run out, an in-app modal asks to sign in; "Continue as guest" dismisses it, "Sign in now" goes to login
Server-enforced	Limits checked before the AI call; rejected sends roll back their bubbles (nothing stored); expired sessions can't post to any account endpoint (<code>ERR_GUEST_NOPE</code>)
Session persistence	Cookie <code>gtoken</code> (HttpOnly, 24 h); chat history kept during the session, survives reload; re-clicking guest mints a fresh session
Clean stats	Guest messages never touch the <code>usage</code> table — admin statistics stay per-real-user
Auto-delete (nightly purge)	Scheduler wipes ALL guest data — sessions, chats, messages — once per day at an admin-configurable local hour picked from a clock-style HH:00 dropdown (default 00:00, <code>guest_purge_hour</code>); plus a manual "Purge guest chat" button (warn-confirm) in Admin → Usage Statistics

Chat Experience

Feature	Description
Smart title	After the first reply, a short 3–6-word title is generated in the background (never overwrites manual renames; guests excluded)
Edit message	Pencil on user bubbles → inline textarea → save = rewrite + truncate everything after, resend automatically

Feature	Description
Regenerate	↺ on AI bubbles truncates from the last user message and resends
Code highlighting	Fenced blocks coloured via bundled highlight.js (v11.9, served from <code>/vendor/</code> , no CDN) + per-block copy button; theme-aware (github-dark/github swap with the UI theme)
Pin / Tag / Fork / Archive	Chat "... " menu — pin keeps chats at top, <code>#tag</code> shown as chip, fork copies the whole conversation as <code>(fork)</code> , archive hides chats into a collapsible "Archived (n)" section

Usage Statistics (Admin)

Compact dashboard: period as segmented pills (1d/7d/30d/90d, default 1d) beside Purge/Reset action buttons. Card row (requests, total/prompt/completion tokens, active users, guest sessions) + a canvas bar chart (no chart library, re-themes light/dark) whose x-axis shows **hour labels (00:00–21:00) for 1d** and **date labels (dd/mm) for 7/30/90d**. Top-10 model & user rankings sit side by side in a 2-column grid with rank numbers, inline bars and token+request meta per row. Endpoints: `GET /api/admin/stats?days=N` (1–365), `POST /api/admin/guest-purge-now`, `PUT /api/admin/reset-usage`.

Reply Stats Strip (v1.0-beta.31)

- **Open-WebUI-style meta chips** under every finished answer: ⌚ generation time (tooltip shows first-token latency), ⚡ output speed (tokens/s), 🗨 context-window usage %, ↑ prompt tokens (sent), ↓ completion tokens (received), ∑ total — each chip has an ID/EN tooltip.
- **Persistent:** stored in `messages.meta` (JSON) → survives refresh/reload, rendered for old replies too, and shown live for temp/guest streams.
- **Context window is AUTO-DETECTED per model** (v1.0-beta.33): the gateway's `/v1/models` metadata (`context_length`, `max_completion_tokens`) feeds the % chip — switch models and the % follows each model's real window. The admin **Fallback context window** setting (validated 1,024-1,048,576, `ERR_BAD_CTXWIN`) is only used for models with no provider metadata (e.g. combo routes).

Models

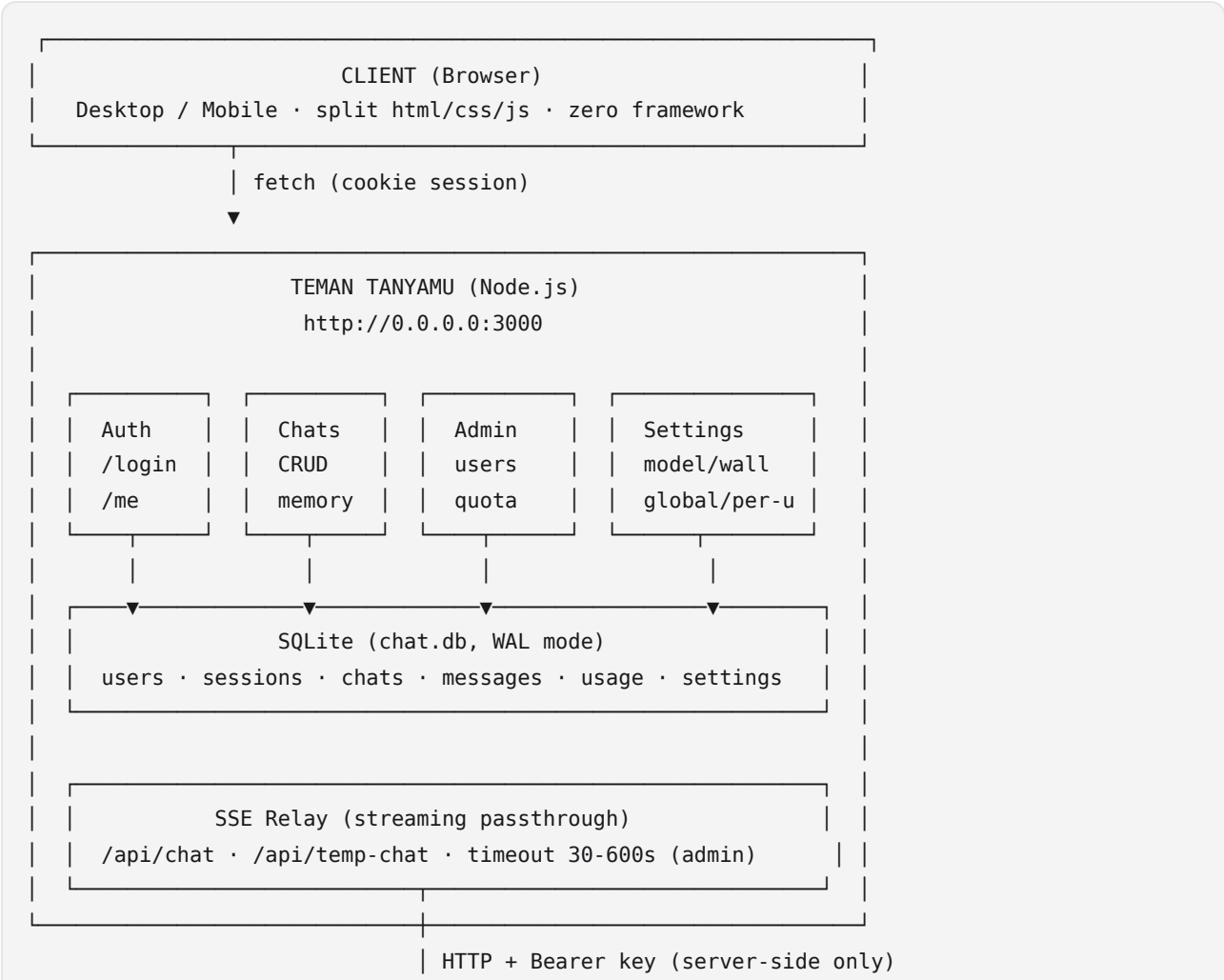
Feature	Description
Model Connection (gateway)	Base URL + API key set in Admin Dashboard (Open WebUI style): masked key display, show/hide eye, Test button (live latency + model count), instant hot-reload, no restart
Auto-fetch	Model list pulled live from 9Router (or any OpenAI-compatible endpoint)
Searchable picker	Custom combobox with instant filter + keyboard nav (↑/↓/Enter/Esc), works for admin & locked users
Global default	Admin setting, applies to all users
Per-user override	Admin sets specific model per user
Per-chat model	Each chat remembers its model
Locked for users	Regular users cannot change model (server-enforced); UI shows a red "Terkunci / Locked" chip

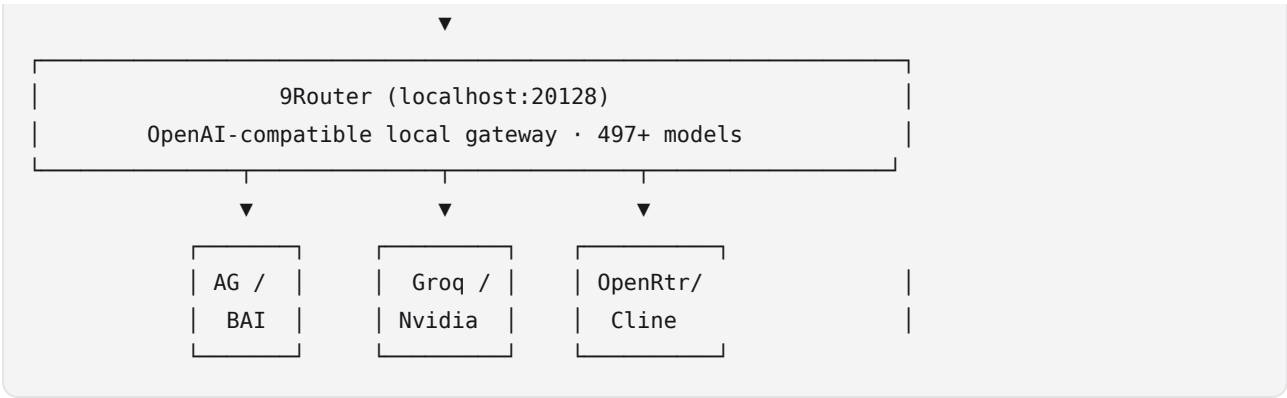
Appearance

Feature	Description
Light / dark theme	Toggle in the sidebar (sun/moon); dark = Open WebUI palette, light derived from it. Persisted per browser; highlight.js CSS theme swaps with it

Feature	Description
Chat wallpaper	6 presets + custom image upload (per-user, optional global); dark overlay only for image wallpapers, gradient presets render as-is
Custom avatar	Auto-downscaled upload, strict base64 validation; rendered in chat bubbles, sidebar and admin list
Global assistant avatar	Admin uploads one AI avatar for everyone (<code>/api/admin/assistant-avatar</code>), or resets to default mask icon
Admin avatar popover	Click any avatar row in the admin user table → upload / reset / cancel (no native prompts anywhere)
Sidebar minimize	Desktop: icon rail (user-row collapses to avatar + stacked actions), state persisted · Mobile: closes the drawer
Bilingual UI	English / Indonesian, all strings translated — dialogs, toasts, admin panel, error codes
Custom dialogs	All confirm/rename/quota dialogs are in-app modals (i18n + consistent style), never native <code>alert/confirm/prompt</code>
Responsive	Mobile drawer sidebar, 44px+ tap targets, admin tables become card stacks, verified 1920→320 px with no horizontal overflow

System Architecture



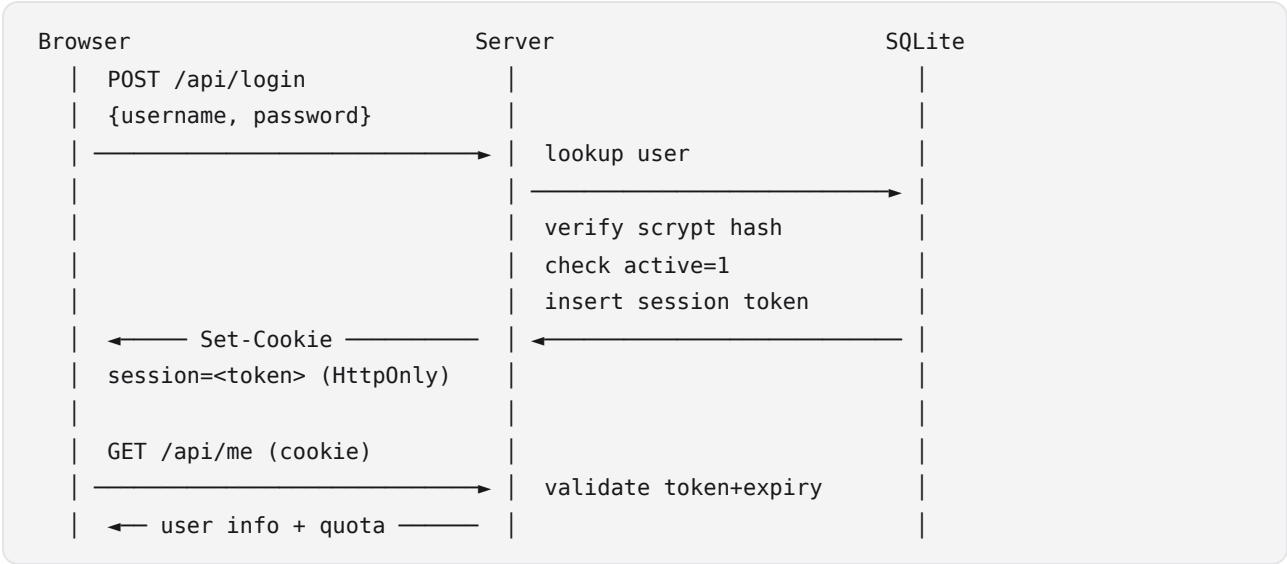


Key Design Decisions

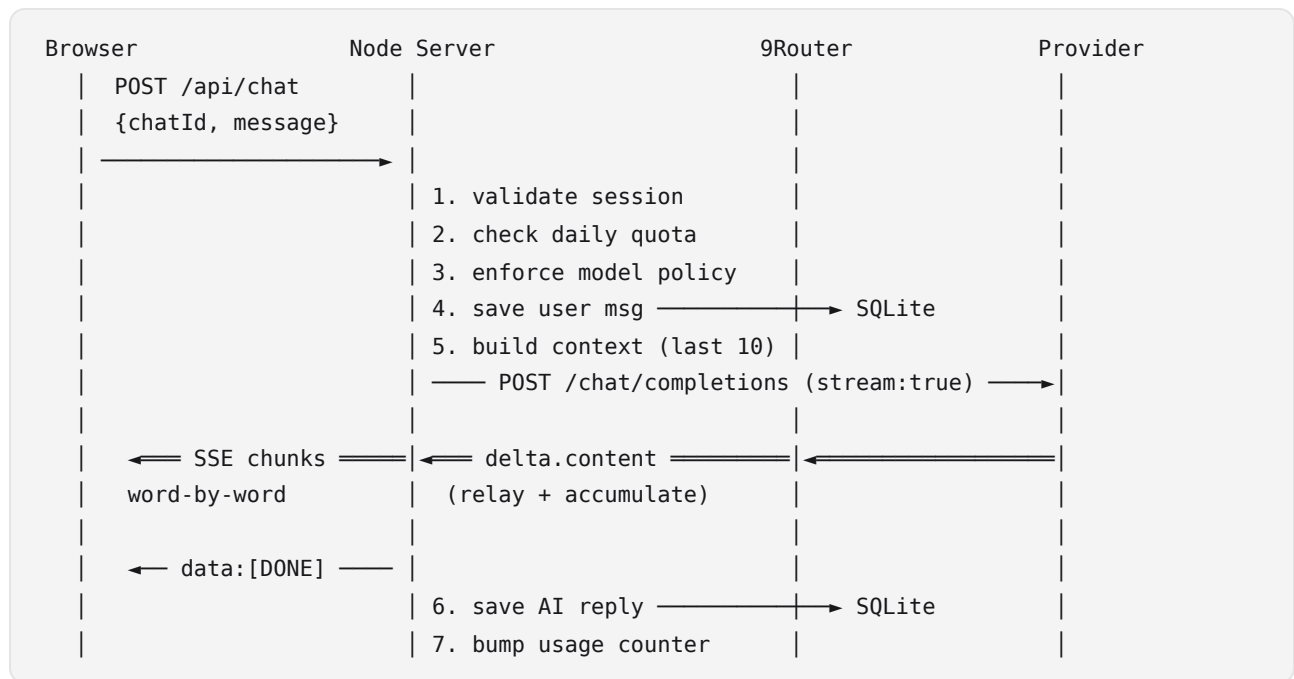
Decision	Reason
Single file backend (<code>server.js</code>)	Easy audit, zero complexity
SQLite over Postgres	Zero-config, WAL mode, single file backup
Server-side API key proxy	Key never reaches the browser
SSE relay (not direct)	Streaming works cross-device, CORS-free
Error codes (<code>ERR_*</code>)	Translated client-side per language
no-cache delivery + versioned assets	HTML: no-store · CSS/JS: no-cache + <code>?v=</code> query (updates always fresh, stale caches bust on release)
Split frontend (<code>style.css</code> + <code>app.js</code>)	Cacheable assets, lintable JS, CSP-ready

Request Flow Diagrams

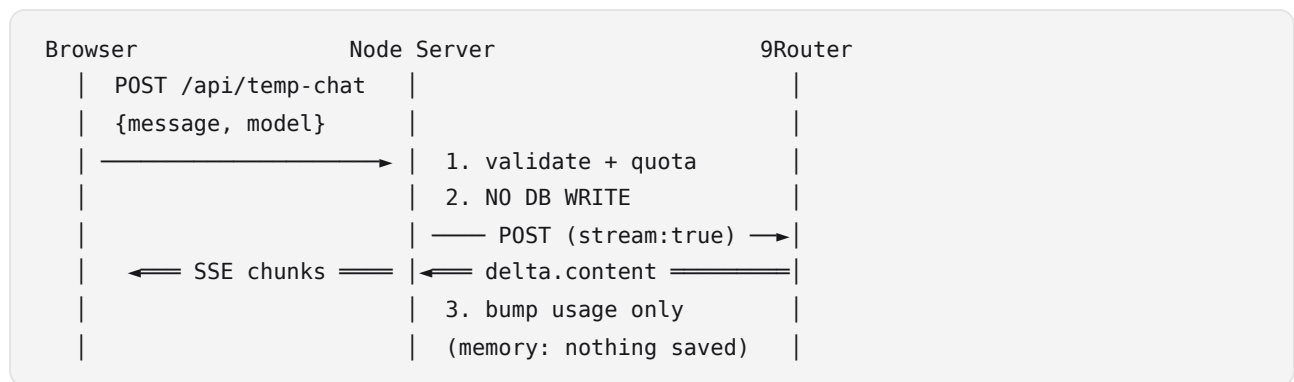
1. Login Flow



2. Streaming Chat Flow



3. Temporary Chat Flow



4. Admin Model Policy

User (non-admin) changes model?

|

└─ UI: selector disabled

└─ Server (double check):

PUT /api/me/model → 403 ERR_ADMIN_ONLY

PATCH /api/chats model → 403 ERR_ADMIN_ONLY

POST /api/chats {model} → forced to resolveUserModel()

POST /api/chat → chat.model auto-corrected

Admin sets global default → settings.default_model

Admin sets user override → users.model_override

Resolution order: user override > global default

Requirements

Component	Minimum
OS	Linux (tested: Ubuntu 24.04)
Node.js	>= 20 (tested: v24.21)
9Router	Running, OpenAI-compatible, default port 20128
Disk	~50 MB (incl. node_modules)

Installation

```
cd dash_ai_me
npm install      # express + better-sqlite3
node server.js  # http://localhost:3000
```

First run creates the admin account automatically:

```
username: admin
password: admin123      # change immediately via UI
```

Management Script (`manage.sh`)

All-in-one service manager (systemd user service):

```
./manage.sh start      # Start service
./manage.sh stop       # Stop service
./manage.sh restart    # Restart service
./manage.sh status     # Status, port, RAM
./manage.sh port <1-65535> # Change port (persistent, auto-restart)
./manage.sh clear-cache # Clean SQLite WAL + npm cache + old journal
./manage.sh enable-boot # Auto-start at boot (enables linger)
./manage.sh disable-boot # Disable boot autostart
./manage.sh logs [n]   # Last n log lines (default 50)
./manage.sh password   # Reset admin password to admin123
./manage.sh help       # Show help
```

`enable-boot` runs `loginctl enable-linger` so the service runs without an active desktop session.

Configuration

Environment Variables

Variable	Default	Description
PORT	3000	Web server port
ROUTER_BASE	–	Boot fallback base URL (used only until set in Admin Dashboard)
ROUTER_KEY	–	Boot fallback API key (same; DB settings take precedence)
ADMIN_PASSWORD	admin123	Initial admin password (first run only)

Example:

```
PORT=8080 ROUTER_BASE=http://192.168.1.10:20128/v1 ROUTER_KEY=*** node server.js
```

Once set via **Admin Dashboard** → **Model Connection**, values in the `settings` table win over env vars — you can change gateway/key anytime from the browser without restart.

File Locations

Path	Description
chat.db	SQLite database (WAL mode)
public/index.html	Struktur halaman SPA (login + app shell)
public/style.css	Seluruh styling frontend
public/app.js	Seluruh logika frontend
server.js	Backend (single file)
.port	Persisted port number (created by <code>manage.sh port</code>)
~/.config/systemd/user/dash-ai-me.service	systemd unit

API Reference

All endpoints require the session cookie except `/api/login` and `/api/health`. Errors return `{"error": "ERR_*"}` codes, translated client-side.

Auth

Method	Endpoint	Body	Description
POST	/api/login	{username, password}	Sign in
POST	/api/logout	-	Destroy session (user + guest cookies)
GET	/api/me	-	User info + quota + effective model + global assistant avatar (guest: guest, quota_used/max, seconds_left, expired)
POST	/api/guest	-	Start/reuse guest session (cookie gtoken ; 403 when guest mode off)

Method	Endpoint	Body	Description
GET	<code>/api/guest-config</code>	-	<code>{enabled}</code> — login screen show/hide for the guest button (no auth)
GET	<code>/api/health</code>	-	Liveness + DB check (no auth, for uptime monitors)

Account

Method	Endpoint	Body	Description
PUT	<code>/api/me/password</code>	<code>{current, next}</code>	Change own password
PUT	<code>/api/me/avatar</code>	<code>{avatar: dataURL null}</code>	Set/remove avatar
PUT	<code>/api/me/model</code>	<code>{model}</code>	Own model override (admin)
DELETE	<code>/api/me/model</code>	-	Clear override (admin)

Chats

Method	Endpoint	Body	Description
GET	<code>/api/chats</code>	-	List own chats
POST	<code>/api/chats</code>	<code>{title?, model?}</code>	Create chat
GET	<code>/api/chats/:id</code>	-	Chat + full messages
PATCH	<code>/api/chats/:id</code>	<code>{title?, model?, pinned?, archived?, tag?}</code>	Rename / lock model (admin) / pin / archive / tag
POST	<code>/api/chats/:id/fork</code>	-	Copy chat + messages into a new <code>(fork)</code> chat
PUT	<code>/api/chats/:id/messages/:msgId</code>	<code>{content}</code>	Edit message content (user messages only)
DELETE	<code>/api/chats/:id/messages/:msgId</code>	-	Truncate chat: delete that message + everything after it
PATCH	<code>/api/chats/:id/lean</code>	<code>{lean: bool}</code>	Toggle lean (no-history) token saver per chat
DELETE	<code>/api/chats/:id</code>	-	Delete chat + messages (auto-revokes share links)
POST	<code>/api/chats/:id/share</code>	-	Create/read-only share token (400 <code>ERR_EMPTY_SHARE</code> on chats with no messages; idempotent)
GET	<code>/api/chats/:id/share</code>	-	Current share token or <code>null</code>
DELETE	<code>/api/chats/:id/share</code>	-	Revoke share link

Public pages (no auth)

Method	Endpoint	Description
GET	<code>/s/:token</code>	Read-only shared-chat HTML page (<code>noindex</code> , <code>no-store</code> , content escaped)
GET	<code>/ {any SPA path}</code>	Unknown non-API paths serve the app (client router: <code>/c/:id</code> , <code>/account</code> , <code>/admin</code>); unknown <code>/api/*</code> stays JSON 404

Memory (cross-chat, per user)

Method	Endpoint	Body	Description
GET	<code>/api/memory</code>	-	List own memory facts + enabled flag
PUT	<code>/api/memory</code>	<code>{facts: string[]}</code>	Replace full fact list (max 20, 240 chars each)
DELETE	<code>/api/memory/:factId</code>	-	Delete one fact
POST	<code>/api/memory/clear</code>	-	Wipe own memory

Chat & Preferences

Method	Endpoint	Body	Description
GET	<code>/api/models</code>	-	Model list from 9Router
POST	<code>/api/chat</code>	<code>{chatId, message}</code>	Streamed chat (SSE); token-budget context + system memory injected; final <code>usage</code> event with prompt/completion split
POST	<code>/api/temp-chat</code>	<code>{message, model?, lean?}</code>	Streamed temp chat (SSE, not saved); <code>lean:true</code> = zero context (no system prompt/memory)
GET	<code>/api/wallpaper</code>	-	Effective wallpaper
PUT	<code>/api/wallpaper</code>	<code>{wallpaper null}</code>	Own wallpaper

Admin

Method	Endpoint	Body	Description
GET	<code>/api/admin/users</code>	-	Users + today's usage + effective model + <code>me</code> id
POST	<code>/api/admin/users</code>	<code>{username, password, role, daily_quota}</code>	Create user
DELETE	<code>/api/admin/users/:id</code>	-	Delete user (self-delete blocked)
PUT	<code>/api/admin/users/:id/password</code>	<code>{password}</code>	Reset password (revokes sessions)
PUT	<code>/api/admin/users/:id/active</code>	<code>{active: bool}</code>	Activate / deactivate (self-deactivate blocked)
PUT	<code>/api/admin/users/:id/quota</code>	<code>{daily_quota}</code>	Edit daily quota (1–999999)

Method	Endpoint	Body	Description
PUT	<code>/api/admin/users/:id/reset-quota</code>	-	Reset one user's daily usage
PUT	<code>/api/admin/reset-usage</code>	-	Reset all users' usage (<code>DELETE FROM usage</code>)
PUT	<code>/api/admin/users/:id/model</code>	<code>{model null}</code>	Per-user model override
POST	<code>/api/admin/avatar</code>	<code>{userId, avatar null}</code>	Upload / reset any user's avatar (self-sync immediate)
POST	<code>/api/admin/assistant-avatar</code>	<code>{avatar null}</code>	Global assistant (AI) avatar for all users
GET	<code>/api/admin/stats</code>	<code>?days=1..365</code> (def 30)	Usage statistics: totals, per-day series, top models/users, guest sessions
POST	<code>/api/admin/guest-purge-now</code>	-	Purge all guest data immediately
GET	<code>/api/admin/settings</code>	-	Global settings + today's token totals (prompt/completion)
PUT	<code>/api/admin/settings</code>	<code>{default_model?, history_token_budget?, max_reply_tokens?, memory_enabled?, timeout_ms?, guest_enabled?, guest_max_chats?, guest_max_minutes?, guest_purge_hour?}</code>	Update settings (budget 200–32000, reply cap 64–8192, timeout 30–600 s, guest chats 1–200, guest minutes 1–60, purge hour 0–23)
GET	<code>/api/admin/router-config</code>	-	Model gateway base URL + masked key + source (<code>database / env</code>)
PUT	<code>/api/admin/router-config</code>	<code>{base_url?, api_key?}</code>	Set gateway base URL & API key (hot-reload, no restart; empty key = keep)
POST	<code>/api/admin/router-test</code>	<code>{base_url?, api_key?}</code>	Probe <code>GET {base}/models</code> with saved or posted (pre-save) creds → latency + model count
PUT	<code>/api/admin/wallpaper</code>	<code>{wallpaper null}</code>	Set global wallpaper

Error Codes

Code	Meaning
<code>ERR_UNAUTH</code>	Not signed in

Code	Meaning
ERR_ADMIN_ONLY	Admin-only action
ERR_FIELDS_REQUIRED	Missing fields
ERR_LOGIN	Wrong credentials
ERR_INACTIVE	Account deactivated
ERR_PW_SHORT	Password under 4 chars
ERR_OLD_PW	Old password wrong
ERR_QUOTA	Daily quota exceeded
ERR_RATE_LIMITED	Too many failed logins: 5 attempts, then 2-minute IP block
ERR_NOT_FOUND	Resource not found
ERR_USERNAME_TAKEN	Username exists
ERR_TOO_BIG	Image too large
ERR_AVATAR_FORMAT	Avatar not an image
ERR_WALLPAPER_FORMAT	Wallpaper format invalid
ERR_SELF_DELETE	Cannot delete own account
ERR_SELF_DEACTIVATE	Cannot deactivate own account
ERR_BAD_BODY	Malformed JSON body
ERR_BAD_BUDGET	History budget out of range (200–32000)
ERR_BAD_MAXTOK	Reply token cap out of range (64–8192)
ERR_BAD_TIMEOUT	Upstream timeout out of range (30–600 s)
ERR_ROUTER_DOWN	9Router unreachable
ERR_ROUTER_NOT_CONFIGURED	Base URL / API key not set yet (Admin Dashboard → Model Connection)
ERR_BASE_REQUIRED	Base URL field empty
ERR_BASE_INVALID	Base URL not a valid http(s) URL
ERR_BASE_BLOCKED_HOST	Base URL host blocked (link-local / metadata addresses)
ERR_TIMEOUT	Upstream model call timed out
ERR_GUEST_DISABLED	Guest mode is turned off
ERR_GUEST_CHATS	Guest message limit reached
ERR_GUEST_TIME	Guest session time limit reached
ERR_GUEST_NOPE	Account feature unavailable in guest mode

Database Schema

```

users      (id, username, password_hash, role, daily_quota,
            model_override, avatar, active, created_at)
-- password_hash: "scrypt1:<salt>:<hash>" (per-user salt);
-- legacy static-salt hex auto-upgrades on next login
sessions   (token, user_id, expires_at)           -- expired rows purged on login
chats      (id, user_id, title, model, lean, pinned, archived, tag,
            summary, created_at, updated_at)      -- lean: 1 = no-history token saver
-- pinned/archived: 0/1; tag: free text (<= 24 chars)
messages   (id, chat_id, role, content, tokens, created_at)
usage      (user_id, date, request_count, tokens_used) -- PK (user_id, date)
shared_chats (token, chat_id, user_id, created_at)
-- token = 16 hex chars, random; read-only public snapshots; revoked on chat delete
guest_sessions (token, guest_id, msgs_used, started_at, expires_at)
-- guest chats stored under user_id = -guest_id; wiped on logout / purge
memories   (id, user_id, content, source, updated_at) -- cross-chat facts (max 20/user)
settings   (key, value)
-- keys: default_model, wallpaper, assistant_avatar, router_base, router_key,
--       user:<id>:wallpaper, history_token_budget, max_reply_tokens,
--       memory_enabled, timeout_ms, memory_sync_chat, memory:last:<id>,
--       guest_enabled, guest_max_chats, guest_max_minutes, guest_purge_hour

```

Access from Other Devices

Server binds to all interfaces. From phones / LAN devices:

http://<laptop-ip>:3000 e.g. http://20.20.100.6:3000

For access over the internet, a Cloudflare Tunnel works out of the box: `trust proxy` is enabled and `cf-connecting-ip` is honored for login rate limiting; the session cookie gets the `Secure` flag automatically when the request arrives via HTTPS.

Performance

Metric	Measured
RAM idle	20-30 MB
RAM peak (streaming)	50-65 MB
CPU idle	~0%
Boot time	< 3 s
Streaming first token	~0.5-2 s (provider dependent)

Security Notes

- Passwords: scrypt with per-user random salt, stored as `scrypt1:<salt>:<hash>`, verified with `crypto.timingSafeEqual`; legacy static-salt hashes upgrade automatically on the next successful login
- Sessions: HttpOnly cookie, SameSite=Lax, 7-day expiry; expired rows purged on login
- **Secure cookie flag**: auto-enabled when behind HTTPS (Cloudflare Tunnel sets `X-Forwarded-Proto`)
- **Login rate limiting**: 5 failed attempts per IP, then a 2-minute block (`ERR_RATE_LIMITED`)
- `trust proxy` enabled: client IP correctly detected behind Cloudflare (`CF-Connecting-IP`)
- **Security headers** on every response: `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: no-referrer`
- **XSS hardening**: avatar & wallpaper validated server-side against strict formats (base64 image types / `linear-gradient(...)` / `#hex` only — no `svg+xml`, no attribute breakout); all user-supplied values escaped at render
- 9Router API key never sent to browser (server-side proxy only)
- Per-user chat isolation enforced in every SQL query
- **Share links**: random 16-hex tokens (unguessable), no auth leakage (server-rendered static HTML, no cookies needed), `noindex` for search engines, `no-store`, chat content HTML-escaped (no script injection); revoke + delete-chat cleanup keep zero orphan rows; empty chats cannot be shared
- **SPA fallback** serves only a constant file path (`public/index.html`) — no user input reaches `sendFile`; API namespace protected with explicit JSON 404 before the catch-all
- Model policy enforced server-side (not just hidden in UI)
- Guest mode: chat/time limits enforced server-side **before** the AI call; guests blocked from all account endpoints (`ERR_GUEST_NOPE`); guest chats isolated under negative user ids and excluded from the `usage` stats table
- Quota checked before every AI call; failed/aborted streams do not consume quota
- Deactivate / password reset revokes all sessions instantly
- Caching: HTML `no-store`, CSS/JS `no-cache` + `?v=` asset version in `index.html` (bump on each release so old client caches invalidate automatically)

Troubleshooting

Problem	Fix
"9Router offline"	Verify gateway: <code>curl http://localhost:20128/v1/models -H "Authorization: Bearer ***"</code> — or click Test in Admin Dashboard → Model Connection
Moved gateway / new key	Admin Dashboard → Model Connection: set Base URL + API key, Test, Save. Applies instantly, no restart
Forgot admin password	<code>./manage.sh password</code>
Port already in use	<code>./manage.sh port 3001</code>
Not auto-starting at boot	<code>./manage.sh enable-boot</code> , verify <code>logincctl show-user \$USER grep Linger</code>
Stale UI after update	Hard refresh: <code>Ctrl+Shift+R</code> (HTML/CSS/JS are all no-cache; rarely needed)
Full reset (nuclear)	<code>./manage.sh stop && rm -f chat.db* && ./manage.sh start</code> — all data gone

Fitur operasional level produksi:

- **Backup otomatis:** tiap boot + harian ke `backups/` (retensi 30 hari), plus tombol "Back up database now" di panel **Sistem & Operasi** pada Admin Dashboard. CLI: `./manage.sh backup` / `restore <file>` / `vacuum`.
- **Audit trail:** login, aksi admin, purge, share, rate-limit tercatat ke tabel `audit` (maks 2000 baris) dan tampil di panel Ops (paginated).
- **Throttle generasi:** default 30 panggilan AI / 90 detik per user (tamu: per IP) → `ERR_GEN_LIMITED`; pembuatan sesi tamu 12/menit/IP. **Bisa disetel live** di Admin → Token Saver: "Anti-spam: maks panggilan AI" (5–1000) + "... per rentang detik" (10–600), tersimpan di settings, langsung berlaku tanpa restart, tercatat di audit trail.
- **Log terstruktur JSON** (`lib/logger.js`) + request log API + endpoint `/api/admin/metrics` (Prometheus text, admin-only).
- **Purge-day persisten:** guard hari purge sekarang disimpan di `settings` (`guest_last_purge_day`), aman restart, pakai tanggal lokal.
- **Regression suite permanen:** `test/e2e.test.js` — 46 assertion (auth, isolasi chat, share+XSS+i18n, guard admin, limiter unit, metrics, backup, purge-route, mode ledger). Jalankan: `./manage.sh test` atau `npm test`.
- **i18n bersama:** `public/i18n.js` dipakai browser DAN server (halaman share ikut bahasa `Accept-Language`).
- **Throttle configurable (v1.0-beta.22)** — Admin → Token Saver: "Max calls" + "per seconds" side-by-side with an inline explainer note (5–1000 / 10–600); stored in settings, applied live without restart, audit-tracked.
- **Motion layer (v1.0-beta.23 → .24)** — chat bubbles rise-in, history & audit rows stagger, stat cards cascade, send button pulses while streaming, login card + footer + form staggered entrance, app↔login cross-fade on login/logout, wrong-password shake, focus glow rings, theme cross-fade, button press feedback — all honoring `prefers-reduced-motion`.
- **Fast chat path (v1.0-beta.24)** — measured login 2–6 s → **0.5 s**: `/api/models` cached 60 s server-side (stale-on-error), boot parallelized (models || chats), post-reply refresh moved off the critical path (non-blocking silent re-render, no animation replay), hidden background LLM tasks (title/summary/memory) queued + delayed 2.5 s so they never steal gateway bandwidth from the next visible message.
- **Never lose a visible reply (v1.0-beta.24)** — if the upstream drops AFTER the answer was streamed & saved, you now get a soft "⚠ interrupted" note instead of the answer being replaced by an error; SSE heartbeat every 10 s prevents idle connection drops; server always emits `[DONE]` so the client never hangs.
- **Ledger mode statistik (v1.0-beta.25)** — tabel baru `usage_events` mencatat SETIAP panggilan AI per (hari, jam, mode, model) — mode normal/eco/temp/guest kini TERHITUNG SEMUA di kartu, chart per-jam, per-model, dan blok baru "Requests by mode" (bar berwarna + keterangan cara catat; historis normal+eco di-seed otomatis dari messages lama). Verified end-to-end: normal ✓ eco ✓ temp ✓ guest ✓. `Reset usage stats` sekarang menyapu kedua ledger (tidak ada ghost chart lagi).
- **Purge guest chat fixed (v1.0-beta.25)** — tombol "Purge guest chat" selalu "failed": rute `/api/admin/guest-purge-now` ternyata terdaftar SETELAH guard 404 SPA-fallback (regresi senyap dari v17) → 404 `ERR_NOT_FOUND`. Blok dipindah sebelum guard; 2 assertion regresi baru ditambahkan ke suite agar bug kelas "route after the 404 wall" tidak pernah balik lagi.
- **Toast anchor aware (v1.0-beta.25)** — popup (purge, save settings, dsb.) kini dipusat ke dialog yang sedang terbuka kalau ada (sebelumnya selalu ke `.main`), jadi terlihat mepet kiri saat panel Admin aktif + sidebar maximize/minimize). Terukur off-center **0px** di kedua posisi sidebar.
- **Poll admin anti-flicker (v1.0-beta.26)** — re-render statistik tiap 5 s dilewati kalau angkanya sama (signature JSON + tema aktif), cascade animasi tidak replay terus-menerus.
- **Long answers fixed (v1.0-beta.27–.29)** — "jawaban terpotong lalu berhenti" caused by two silent killers: total 120 s hard timeout (killed healthy long streams) and the 1024-token reply cap. Timeout is now an *idle* watchdog (bytes flowing = alive; only dead connections abort), max-reply default raised to 4096 (slider up to 16384, live-apply), and a reply that still hits the cap shows a soft hint "⚠ ...max reply..." instead of dying quietly. Verified end-

to-end: the user's exact question "sejarah indonesia abad 15 - hari ini" streams 3 718 chars / 3 346 completion tokens, clean [DONE].

- **Composer border regression (v1.0-beta.27)** — the v23 global focus-glow leaked into the chat textarea and drew a ring around it; `#input-msg:focus` is explicitly borderless again, the composer keeps its subtle `:focus-within` highlight only. Measured: border 0px, shadow none, while typing.
- **"Idle timeout" labels** — Timeout setting renamed (EN/ID) so admins understand it no longer limits total answer length/time, only silent-stall detection.
- **HOTFIX fatal (v1.0-beta.30)** — v29 left a `ReferenceError` in the client stream loop (`aborted` , a SERVER-side variable, leaked into `send()`): every AI chat died on the first chunk with "aborted is not defined", output invisible until refresh (the server finished & saved anyway). Loop now breaks on `done` only; static regression assertion added to the suite (47 total) so server-only identifiers in client reader loops can never ship again.
- **Reply Stats Strip (v1.0-beta.31-32)** — Open-WebUI-style meta chips under every finished answer (time, tok/s, context %, $\uparrow$ prompt / $\downarrow$ completion / Σ total tokens) with ID/EN tooltips; persisted in the new `messages.meta` column so they survive refresh; live for temp & guest streams too; admin Token Saver gains a validated **Model context window** setting (`ERR_BAD_CTXWIN`). Suite now 50.
- **Meta chip arrows swapped (v1.0-beta.32)** — $\uparrow$ = prompt sent, $\downarrow$ = reply received (chat/network convention everyone reads intuitively; the old well-metaphor read backwards to users).
- **Context window AUTO-DETECT per model (v1.0-beta.33)** — gateway `/v1/models` metadata (`context_length` , `max_completion_tokens`) is cached with the model list (plus an 8s-after-boot warm fetch) and feeds the  chip live: each reply's % uses the window of the model that generated it (gemini-3.5-flash $\rightarrow$ 1.049k $\rightarrow$ 6.2k prompt = 1%, not the scary 50% of the old fixed 8k). Models without metadata (e.g. `auto-change` combo) fall back to the admin setting, now honestly labelled **Fallback context window**; tooltip also shows the model's max output. Verified end-to-end on both paths; temp/guest streams included.
- **Versi konsisten:** `package.json` `1.0.0-beta.33` = `APP_VERSION` server = cache-buster aset; git tag per rilis.

Changelog

v1.0-beta (current)

- **Deep-link URLs** — click a chat title for `/c/:id` (address bar stays clean otherwise); `/account` & `/admin` deep links; browser back/refresh/bookmark work (SPA fallback; `/api/*` 404 JSON)
- **Stats polish** — segmented period pills (1/7/30/90d) with hour/date chart axis labels, 2-column ranked model+user panels, purge-hour as a clock-style HH:00 dropdown
- **Share links** — read-only public chat pages at `/s/:token` (create/revoke from chat $\cdots$ menu, clipboard copy, `noindex` , XSS-safe snapshot; empty chats blocked; deleting a chat revokes links)
- **Chat Experience** — smart auto-title after first reply, inline edit & regenerate (truncate + resend), highlight.js code blocks with copy button, pin / tag / fork / archive chat menu
- **Light / dark theme** — sidebar toggle, persisted per browser, hljs + stats chart re-theme live
- **Usage Statistics (admin)** — 1/7/30/90-day cards, canvas daily-token chart, top models/users bars, purge-now + reset-stats buttons
- **Scheduled guest purge** — wipes all guest data daily at an admin-set hour (`guest_purge_hour` , default 00:00) + manual "Purge guest chats now" button
- **Guest Mode** — login-free access via a "Try as guest" button; per-guest message limit (1–200) and session time limit (1–60 min) configurable in Admin Dashboard; live countdown chip; in-app login popup on chats/time

exhaustion; server-enforced quota + endpoint lockdown; bilingual (EN/ID); guest data purged on logout

- **Model Connection in Admin Dashboard** — gateway Base URL + API key stored in DB (no more hardcode), masked key + eye toggle, live Test button (latency + model count), hot-reload, SSRF guard on base URL
- Searchable model combobox, per-user quota edit, realtime quota UI (post-reply + 5 s admin poll), global assistant avatar, custom dialogs (confirm/rename/quota), red model-lock chip, minimized sidebar fixes, versioned asset cache-busting